

# On the Nature of Reactance and its Role in Persuasive Health Communication

James Price Dillard & Lijiang Shen

*Reactance theory might be profitably applied to understanding failures in persuasive health communication but for one drawback: The developer of the theory contends that reactance cannot be measured. Rejecting this position, this paper develops four alternative conceptual perspectives on the nature of reactance (i.e., combinations of cognition and affect), then provides an empirical test of each. Two parallel studies were conducted, one advocating flossing ( $N = 196$ ), the other urging students to limit their alcohol intake ( $N = 200$ ). In both cases, a composite index of anger and negative cognitions fully mediated the effects of threat-to-freedom and trait reactance on attitude and intention. The data showed that, in fact, reactance can be operationalized as a composite of self-report indices of anger and negative cognitions. The implications for persuasive communication, in general, are considered as well the specific findings for flossing and drinking.*

**Keywords:** Reactance; Anger; Threat to Freedom; Persuasion

Persuasive attempts of all sorts, including public health campaigns, often fail to produce the desired effect (Foxcraft, Lister-Sharp, & Lowe, 1997; Wallack, 1981; Wilde, 1993). In fact, in some cases, they produce results directly at odds with their intent (Guttman, Kegler, & McLeroy, 1996; Hornik, 2002; Stewart & Martin, 1994). The theory of psychological reactance (Brehm, 1966; Brehm & Brehm, 1981; Wicklund, 1974) provides one theoretical perspective through which these suatory

---

James Dillard, Department of Communication Arts & Sciences, Pennsylvania State University, and Lijiang Shen, Center for Communication Research, University of Wisconsin-Madison, Madison, Wisconsin, USA. We thank Mike Stephenson for his reactions to an earlier version of this manuscript. Correspondence to: James Dillard, Department of Communication Arts & Sciences, Sparks Building, Pennsylvania State University, University Park, PA 16802, USA. Email: jpd16@psu.edu

miscarriages might be understood. The theory contends that any persuasive message may arouse a motivation to reject the advocacy. That motivation is called reactance. From its inception to the present, the theory has been called upon to explain resistance to persuasion (Buller, Borland, & Burgoon, 1998; Burgoon, Alvaro, Grandpre, & Voulodakis, 2002; Ringold, 2002).

The primary limiting factor in the application of reactance theory to persuasive campaigns is the ephemeral nature of its central, explanatory construct. The creator of the theory contends that reactance cannot be measured (Brehm, 1966; Brehm & Brehm, 1981). Although that claim may have been accurate during the infancy of reactance theory, advances in the study of persuasion since that time suggest that it should be reconsidered.

The rationale for this project is presented in four parts. First, we provide a brief overview of the theory of psychological reactance. Second, we examine the notion of reactance itself in more detail. In part, this involves reconceptualizing reactance along more contemporary lines. Third, we consider a subset of the many variables that are likely to lead to reactance arousal and propose a corresponding series of hypotheses. The general supposition is that the effect of these variables on attitude is mediated by one of the alternative conceptualizations of reactance. Fourth, two studies, designed to test that supposition, form the empirical core of this paper. These sections each speak to the larger themes that motivated the research. That is, what is the nature of reactance and what role does it play in persuasive health communication?

### **The Theory of Psychological Reactance**

There are four essential elements to reactance theory: Freedom, threat to freedom, reactance, and restoration of freedom. The notion of freedom is not freedom in general terms; it is “not abstract considerations, but concrete behavioral realities” (Brehm & Brehm, 1981, p. 12). However, the concept of free behaviors is defined broadly so as to include actions, as well as emotions and attitudes (Brehm, 1966; Wicklund, 1974). Individuals possess freedoms only to the extent that they have knowledge of them and perceive that they are capable of enacting the behavior.

Given that an individual perceives a specific freedom, any force on the individual that makes it more difficult for him or her to exercise that freedom constitutes a threat (Brehm, 1966; Brehm & Brehm, 1981). Even an impersonal event, such as the weather, can be viewed as a threat, if it renders more difficult the exercise of a freedom. However, social influence as a threat is most pertinent to questions of persuasive health communication. In fact, one of the basic claims of the theory is that high-pressure communicators are likely to be seen as threats to freedom (Brehm & Brehm, 1981; Wicklund, 1974).

Psychological reactance is “the motivational state that is hypothesized to occur when a freedom is eliminated or threatened with elimination” (Brehm & Brehm, 1981, p. 37). The theory contends that when a perceived freedom is eliminated or threatened with elimination, the individual will be motivated to reestablish that freedom. *Direct restoration* of the freedom involves doing the forbidden act. In addition,

freedoms may be restored indirectly *by increasing liking for the threatened choice* (Brehm, Stires, Sensenig, & Shaban, 1966; Hammock & Brehm, 1966), *derogating the source of threat* (Kohn & Barnes, 1977; Schwarz, Frey, & Kumpf, 1980; Smith, 1977; Worchel, 1974), denying the existence of the threat (Worchel & Andreoli, 1974; Worchel, Andreoli, & Archer, 1976), or *by exercising a different freedom* to gain feeling of control and choice (Wicklund, 1974). Although all of these means for reducing reactance have been the focus of research, reduced or boomerang attitude change has captured the lion's share of attention.

### The Nature of Reactance

To clarify the role that reactance might play in the processing of health messages, it is useful to examine the construct from both a conceptual and an operational standpoint. Conceptually speaking, reactance has been defined primarily in terms of its antecedents and outcomes. For example, as expressed in the quotation above, the proximal cause of reactance is a perceived threat to freedom: Reactance is "the motivational state that is hypothesized to occur when a freedom is eliminated or threatened with elimination" (Brehm & Brehm, 1981, p. 37). Thus, reactance is the force that prompts certain outcomes, particularly, efforts to re-establish the threatened freedom by either direct or indirect means. Apart from a brief mention of the possibility that individuals "may be aware of hostile and aggressive feelings" (Brehm, 1966, p. 9), if the level of reactance arousal is high, the nature of reactance itself remains remarkably underdetermined. Brehm's apparent reluctance to provide greater conceptual explication of the principal mechanism of the theory may arise from his belief concerning the potential for measurement of it. According to Brehm and Brehm (1981), "reactance has the status of an intervening, hypothetical variable . . . We *cannot* measure reactance directly, but hypothesizing its existence allows us to predict a variety of behavioral effects" (p. 37, our emphasis). If one of the purposes of a conceptual definition is to provide guidance on operationalization (Chaffee, 1991), such guidance becomes superfluous when operationalization is viewed as impossible.

Since the time of Brehm's writings, scholars have applied ideas drawn from reactance theory to a number of different domains. And, in the course of analyzing and extending the theory, reactance has been implicitly and explicitly defined in several different ways. Building on work of earlier authors, we discern four distinct means of characterizing reactance. In the first, reactance is viewed as purely cognitive. Research in the cognitive response tradition clearly adopts this perspective (e.g., Petty & Cacioppo, 1986), as does work on the clinical manifestations of reactance (Kelly & Nauta, 1997). One advantage to conceiving of reactance in this way is that it immediately becomes measurable through a variety of self-report techniques. The means that is most obviously relevant to questions of persuasion is the widely used thought-listing technique (Cacioppo & Petty, 1981). This purely cognitive view suggests that reactance can be conceived of and operationalized as counter-arguing.

Second, citing similarities between antecedents of reactance and cognitive appraisals that lead to anger, some writers suggest that reactance might be considered, in whole

or in part, as an emotion (Dillard & Meijnders, 2002; Nabi, 2002). Certainly, this claim aligns well with Brehm's description of reactance as the experience of hostile and aggressive feelings (Seltzer, 1983; White & Zimbardo, 1980; Wicklund, 1974). In this view then, reactance might be considered more or less synonymous with the family of concepts that index varying degrees of anger (e.g., irritation, annoyance, and rage). From this perspective, reactance might be operationalized in various ways including asking individuals to make a judgment on a close-ended scale regarding the degree to which they are experiencing anger.

A third logical option holds that reactance might be considered as both affect *and* cognition. Though unrelated to reactance *per se*, one example of this type of thinking can be seen in Leventhal's (1970) parallel processing model. He posits that individuals have both cognitive and emotional reactions to persuasive health messages and that those reactions have unique effects on message acceptance. Evidence consistent with both points can be found in studies of cognitive and emotional responses to public service announcements (Dillard & Peck, 2000, 2001; Dillard, Plotnick, Godbold, Freimuth, & Edgar, 1996; Stephenson, 2003; Witte, 1994).

The final possibility also suggests that reactance has both cognitive and affective components. However, unlike the previous position, which specified distinct effects, in this fourth perspective cognition and affect are intertwined. In fact, they are intertwined to such a degree that their effects on persuasion cannot be disentangled. Such a view is most compatible with a conception of motivation as an alloy of its components, rather than a simple sum of distinct elements (as is implied by the previous position). Considered in unison, these four conceptions of reactance can be expressed as a pair of research questions:

RQ1: Should reactance be conceptualized and operationalized as cognition, affect, or both?

RQ2: If reactance has both cognitive and affective components, how are the two components combined?

We should note some potential objections to thinking of reactance as either counter-arguing or as anger. One reading of the theory suggests that reactance cannot be construed as isomorphic with counter-arguing (or anger) because it is the cause of counter-arguing (or anger). The theory might be interpreted this way, but to do so immediately returns us to the circumstance in which reactance cannot be measured<sup>1</sup>. We see that as undesirable.

Another possible objection concerns the fact that counter-arguing (or anger) might be prompted by message processing goals (e.g., defensive processing) or aspects of a message (e.g., weak arguments). As this argument goes, if our proposed mediators are responsive to other variables than those detailed by Brehm (1966; Brehm & Brehm, 1981), then they are not unique. And, if they are not unique, they should not be considered reactance. However, this argument overlooks the fact that all motivations implicate cognition, affect, or both (e.g., Kuhl, 1986). That is, whether the motivation is one of achievement, self-actualization, reactance, and so on, it will guide attention, influence thought processes (Derryberry & Tucker, 1994), stimulate feelings, and direct behavior (Kuhl, 1986; Lang, 2000). Accordingly, we are inclined to give little weight to this position.

### Reactance as a Mediator

Although one might wish for a more detailed explication of reactance itself, the theory is quite explicit concerning the role of reactance in the persuasion process: Reactance is the state that mediates the effects of threat to freedom on various outcomes such as attitude and behavior. This point is of signal importance insofar as it suggests a strategy for addressing the two research questions posed above. If it were possible to identify variables that were accepted causes and consequences of reactance, then one could test the various combinations of cognition and affect for their ability to mediate the antecedent-consequent relationship. To the extent that mediation was observed, it would be reasonable to claim that one or more of those combinations were functionally synonymous with reactance. To implement that strategy we turn next to a brief and highly selective review of the precursors of reactance.

### Two Antecedents of Reactance

#### *Strength of the Threat to Freedom*

The idea that threats to freedom vary in strength is broad enough to imply that perceptions of threat are likely to be responsive to multiple message features. Given the goals of this study, our aim was not to attempt to advance understanding of the impact of specific message features on reactance. Rather, we sought (a) to create a strong manipulation that (b) could be unequivocally judged as having construct validity by anyone familiar with the theory. A review of the literature suggested that these aims could be achieved by purposefully confounding *intent to persuade* and *language intensity*.

Several studies demonstrate that manipulations of perceived intent to persuade produces results that are consistent with the theoretical predictions of reactance theory (Heller, Pallak, & Picek, 1973; Kohn & Barnes, 1977; Worchel & Brehm, 1970). Strong evidence of the negative impact of intent on persuasion can be found in Benoit's (1998) meta-analysis, which reported a homogeneous effect across 12 investigations.

Other research indicates that intense, forceful, or dogmatic language may increase the magnitude of reactance (Bensley & Wu, 1991; Doob & Zabrack, 1971; Worchel & Brehm, 1970). For example, M. Brehm (reported in Brehm, 1966) conducted a study in which students at the University of Kansas were exposed to a message that argued for a stronger program in intercollegiate athletics. Students who received the version of the message that ended with the sentence "You, as college students, must inevitably draw the same conclusion" showed less agreement with the message than those for whom the sentence was omitted. More recent work, conducted by Buller and his colleagues (Buller et al., 1998, 2000), shows similar effects with some qualifications involving message format and prior behavioral intentions. Overall, theory and previous findings led us to predict that:

H1: Strength of the threat to freedom is directly correlated with magnitude of reactance.

#### *Trait Reactance Proneness*

Although reactance was initially conceived as situation specific (Brehm, 1966; Wicklund, 1974), Brehm and Brehm (1981) recognized that individuals may vary in

their trait proneness to reactance arousal. Subsequently, several scholars developed scales to measure persons' transituational propensity to experience reactance (Dowd, Hughs, Brockbank, & Halpain, 1988; Hong, 1992; Hong & Faedda, 1996; Hong & Page, 1989; Merz, 1983). Of the available options, Hong's scale shows the best conceptual correspondence with the reactance construct and the most favorable psychometric properties. In line with the simple notion that persons high in trait reactance should be more prone to experience reactance in response to a persuasive message, we hypothesized that:

H2: Trait reactance proneness is directly correlated with magnitude of reactance.

### *The Combined Effects of Threat and Proneness*

It is difficult to derive an unequivocal hypothesis regarding the combined impact of threat and trait proneness on the magnitude of reactance. To be sure, main effects-only findings would pose no threat to the theory. However, an interaction between the two antecedents might also be viewed as compatible with the theory assuming that it was ordinal in form. For example, an interaction in which person high in trait proneness showed greater reactivity to a strong threat than weak threat seems quite plausible. Still, it is not clear that the theory demands such a pattern of data. Lacking guidance from the theory, we asked simply:

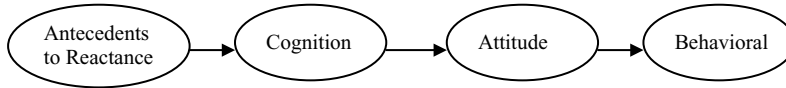
RQ3: Is there an interaction between threat and reactance proneness on magnitude of reactance arousal? And, if so, what is the form of that interaction?

### **Modeling the Reactance Process**

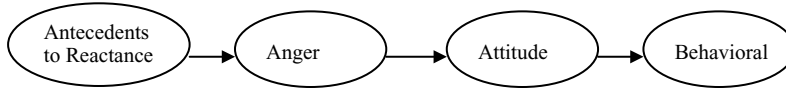
As noted earlier, reactance theory specifies a process in which antecedents of reactance bring about reactance, which, in turn, prompts efforts to restore the threatened freedom. Given four possible conceptions of reactance, four corresponding processes are possible. An illustration of each is given in Figure 1. Because the first model assumes that reactance is a purely cognitive phenomenon, we label it the Single Process Cognitive Model. For parallel reasons, the second model is termed the Single Process Affective Model. The third option, which assumes that cognition and affect can be discriminated, is the Dual Process Model. The Intertwined Process Model suggests that affect and cognition are so closely interwoven that they are better thought of as indicators of an underlying concept than as distinct phenomena.

To the process specified by Brehm (1966), we have added a causal link from attitude to behavioral intention. This prediction, drawn from the theories of reasoned action (Fishbein & Ajzen, 1975) and planned behavior (Ajzen, 1991), has been supported in numerous studies (Kim & Hunter, 1993a, 1993b). The purpose in including it here is to add a degree of complexity to the models under study. To do so increases the difficulty of fitting the models to the data and correspondingly enhances the degree of confidence that we might reasonably place in any of the models that successfully reproduce the data. In addition, it extends the reach of reactance theory from attitude to behavioral intention.

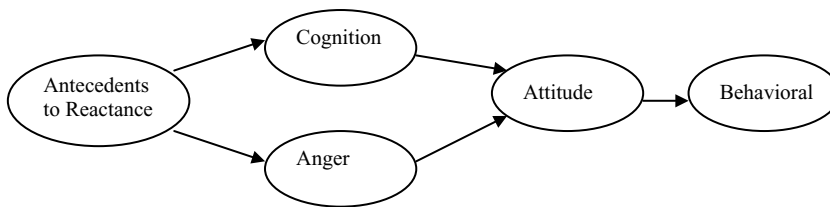
*A Single Process Cognitive Model*



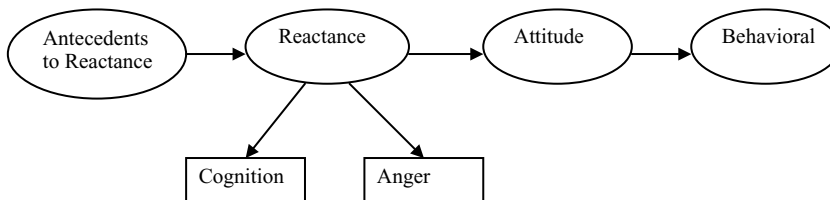
*A Single Process Affective Model*



*A Dual Process Cognitive-affective Model*



*An Intertwined Process Cognitive-affective Model*



**Figure 1** Comparison of four models of reactance.

## Method

Two studies were conducted that were conceptual replications of one another. In each case, respondents read one of two versions of a persuasive message that varied in strength of threat, then provided data on their cognitive and affective responses, as well as their attitude, behavioral intention, and trait reactance proneness. The studies differed only in message topic: flossing versus binge drinking. These two topics were chosen to capture a variety of differences and thereby increase the generalizability of the results. Specifically, one message promoted a private action (i.e., flossing), while the other advocated reducing a public behavior (i.e., drinking).

## Participants

All participants were recruited from undergraduate classes in Communication or Journalism at the University of Wisconsin-Madison. They received a small portion of



extra credit for participating in the study. The 205 subjects in the drinking study were between ages 18 and 30 ( $M = 20.79$ ,  $SD = 1.46$ ). White/Caucasian subjects accounted for 90.2% of the sample, Asian descent for 5.9%, Hispanic descent for 2.4%, and other, including African descent, 1.5%. The sample was 69.3% female and 30.7% male.

The 202 participants in the flossing study were between ages 18 and 32 ( $M = 20.64$ ,  $SD = 1.43$ ). White/Caucasian subjects accounted for 88.6% of the sample, Asian descent for 6.9%, Hispanic descent for 3.0%, and other, including African descent, 1.5%. This sample was 72.3% female and 27.7% male. List-wise deletion reduced the  $N$ s to 196 (flossing) and 200 (drinking).

### *Message Design*

Both messages followed the standard format for a fear appeal in that they consisted of a threat-to-health component and an action or recommendation component (Rogers, 1983). The threat-to-health portion of the messages discussed the negative consequences of not flossing and binge drinking respectively. A summary of the arguments is given in Table 1.

The strength of threat-to-freedom manipulation appeared in the action component of each message. In each case, forceful language was used in the strong threat condition and milder, more polite terms were chosen for the weak threat condition. Tables 2 and 3 present the exact wording for each of the conditions in each message.

The anti-drinking appeal was a bit longer than the flossing message (1167 words versus 1025). The appeals also differed in that the flossing message contained a color picture of affected gums and a paragraph explaining how to floss just prior to the action component (i.e., the threat-to-freedom manipulation). In all other respects, the messages were fairly similar. Both were formatted on multiple pages and presented to participants in small plastic notebooks.

### *Procedure*

When the participants arrived at the lab, they were told that they would be evaluating some informational messages on a health topic that was relevant to college students. They were assigned to either the high or low threat-to-freedom condition by intermingling the two versions of the message booklets. Due to the sensitivity of the binge drinking topic and the fact that most of the participants were underage, anonymity was assured and achieved. First, subjects signed and dated the consent forms, which were removed from the questionnaires, and passed to the experimenter face down. Next, they were instructed to turn to a pre-message questionnaire that contained a variety of items not relevant to the current report. At this point, participants read the threat-to-health component of the message and provided data on their reactions to it. When this was completed, they read the recommendation component of the message, which contained the threat-to-freedom manipulation, and provided data on their cognitive and affective responses. Only the cognitive and affective reactions to the manipulation were used in subsequent analyses. Finally, participants filled out a questionnaire that contained outcome measures such as manipulation check, attitude toward the behavior, and behavioral intention, as well as



**Table 1** Summary of the Threat-to-health Portions of the Messages

| The flossing message                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | The binge drinking message                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. Due to gum disease, the gums and bone that support the teeth can become painfully and irreversibly damaged. Left unattended, teeth will eventually loosen and fall out.</li> <li>2. Up to 30% of the population may be genetically susceptible to gum disease. If you are in this group, you are up to six times more likely to develop some form of periodontal disease.</li> <li>3. There are other consequences of gum disease: Bad breath, stroke, and pneumonia.</li> <li>4. Instructions on how to floss.</li> </ol> | <ol style="list-style-type: none"> <li>1. Over-consumption is a big problem in 2001. If you are a student at this university, there is a two-thirds chance that you are a binge drinker.</li> <li>2. There is clear evidence that alcohol consumption can reduce your academic performance.</li> <li>3. Too much alcohol increases the chance that you will become a victim or an assailant in a forced sex situation. It also increases the risk of STD.</li> <li>4. Binge drinking not only hurts your body, it also affects the quality of life on campus and it is everyone's problem.</li> </ol> |

Hong's reactance scale. The whole process took approximately 40 minutes. Participants were thanked and their questions answered before they left the lab.

### Measures

For all multi-item measures, principal axis factor analyses followed by varimax rotation were employed to assess dimensionality. In every case, the scales appeared to be unidimensional.

**Table 2** Threat-to-freedom Manipulation in the Flossing Message

| High threat                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Low threat                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regular flossing can greatly reduce the chances of gum disease. As any sensible person can see, there is really no choice when it comes to flossing: You simply have to do it. In fact, the scientific evidence showing a link between gum disease and failure to floss is so overwhelming that only a fool would possibly argue with it. And the fact that gum disease can become the basis for other severe problems such as stroke and pneumonia makes it just stupid not to floss every single day of your life. | Regular flossing can greatly reduce the chances of gum disease. And most people would agree that flossing is worthy of serious consideration. There is pretty compelling evidence showing a link between gum disease and the failure to floss. The majority of dentists view that evidence as strong and clear. And, the fact that gum disease can lead to other severe problems such as heart disease, stroke, diabetes, and pneumonia means that you might want to think about making flossing a regular habit. |
| So, if you floss already, don't stop even for a day. And, if you haven't been flossing, right now is the time to start. Today. Do it because you have to. Floss every single day.                                                                                                                                                                                                                                                                                                                                    | So, if you floss already, keep up the good work. And if you haven't been flossing, now might be a good time to start. In fact, you may want to try it today.                                                                                                                                                                                                                                                                                                                                                      |
| Flossing: It's easy. Do it because you have to! Set a goal for yourself to floss everyday during the next week (starting today)!                                                                                                                                                                                                                                                                                                                                                                                     | Flossing: It's easy. Why not give it a try? Set a goal for yourself to floss everyday during the next week (starting today)!                                                                                                                                                                                                                                                                                                                                                                                      |

**Table 3** Threat-to-freedom Manipulation in the Drinking Message

| High threat                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Low threat                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>Responsible Drinking: You Have to Do It</i></p> <p>The previous pages make it crystal clear: There is unequivocal evidence that over-consumption of alcohol is implicated in reduced school performance, sexual violence, secondary effects on others and physical harm to the drinker. In fact, any reasonable person has to agree that over-consumption of alcohol is a serious campus problem that demands immediate attention. No other conclusion makes any sense. Stop the denial. There is a problem and you have to be part of the solution.</p> <p>So if you drink, drink responsibly. Three drinks is a safe, reasonable, and responsible limit and it's the limit that you need to stick to. Do it.</p> | <p><i>Consider Responsible Drinking</i></p> <p>As the previous pages tried to show, there is pretty compelling evidence that over-consumption of alcohol is implicated in reduced school performance, sexual violence, secondary effects on others and physical harm to the drinker. In fact, most people agree that over-consumption of alcohol is a serious campus problem that needs to be addressed right away. It's a sensible conclusion and one that is hard to deny. There is a problem and you have a chance to be part of the solution.</p> <p>So if you drink, think about drinking responsibly. Perhaps three drinks is a safe, reasonable, and responsible limit and it's a limit you can live with. Why not give responsible drinking a try?</p> |

#### *Induction check*

Perceived threat to freedom was measured by four Likert-type scale items: "The message threatened my freedom to choose," "The message tried to make a decision for me," "The message tried to manipulate me," and "The message tried to pressure me." Participants were presented with a 5-point response scale in which 1 = strongly disagree, 2 = disagree, 3 = neutral/don't know, 4 = agree, and 5 = strongly agree. The alpha reliability of the scale was .83 in the flossing data and .87 in the drinking data.

#### *Anger*

Anger was measured using four items that had been validated in previous studies: irritated, angry, annoyed, and aggravated (Dillard & Peck, 2000; Dillard et al., 1996). The 5-point response scale was anchored at 0 = "none of this feeling" and 4 = "a great deal of this feeling." Alpha reliability was .92 in the flossing data and .94 in the alcohol data.

#### *Cognitive responses*

Participants were asked to write out whatever was in their minds when they finished reading the action component of the message. The resulting data were coded in a four-step sequence by three coders working in inter-locking pairs. First, the coders segmented the data into psychological thought units. Agreement among the three pairs of coders was 97%, 96%, and 95%.

Second, because we viewed open-ended reports of affect as redundant with the close-ended reports, affective responses were identified and removed. To assist with this step, coders relied on a list of feeling terms compiled by Shaver, Schartz, Kirson,

& O'Connor (1987). A unit was classified as affective whenever those words appeared and cognitive otherwise ( $\kappa = 1.00, 1.00, \text{ and } 1.00$ ).

Third, coders evaluated whether or not the cognitive responses were relevant to the message. The purpose of this step is to eliminate irrelevant cognitions and thereby reduce the level of noise in the data ( $\kappa = 0.94, 0.91, \text{ and } 0.89$ ).

Finally, the remaining data were coded either as (a) supportive thoughts, (b) neutral thoughts, or (c) negative thoughts. Supportive thoughts were defined as responses that expressed agreement with the message, self-identification, and positive thoughts toward the message, the message source, or the advocacy; and intention to comply with the advocacy in the message, etc. Negative thoughts were defined as responses that expressed disagreement with the message, negative intention to comply with the advocacy, intention to engage in the risky behavior, derogations of the source, etc. Neutral thoughts were defined as non-evaluative responses to the message, such as "This message is printed on blue paper" ( $\kappa = 0.76, 0.74, \text{ and } 0.70$ ). Only the negative cognitions were used in subsequent data analyses. In the flossing data, participants generated an average of 1.47 negative thoughts ( $SD = 1.85$ ) in response to the action component of the message. The corresponding numbers for the drinking data were 1.47 ( $SD = 1.76$ ).

#### *Attitude*

Participants' attitudes toward the message advocacy (i.e., "Flossing regularly is" and "To limit one's alcohol consumption to three drinks or less is") were measured by seven 7-point semantic differential questions. The word pairs used were: bad/good; foolish/wise; unfavorable/favorable; negative/positive; undesirable/desirable; unnecessary/necessary; and detrimental/beneficial. Alpha reliabilities were .84 in the flossing data and .89 in the drinking data.

#### *Behavioral intention*

Behavioral intention was measured by a 100-point, single-item estimate of the likelihood that participants would floss regularly in the following week or limit their alcohol consumption to three drinks in the following week.

#### *Reactance proneness*

Hong's (1992; Hong & Faedda, 1996; Hong & Page, 1989) reactance scale was used to measure trait reactance proneness. Due to instability of the scale in some prior studies (see Hong, 1992, vs. Hong & Faedda, 1996), a series of factor analyses were conducted to evaluate the dimensionality of the scale. The results indicated first-order multidimensionality, but second-order uni-dimensionality. The details of these analyses are presented in Shen and Dillard (in press). Thus, we treated the set of items as a single measure of trait reactance proneness. The alpha reliabilities in the flossing and drinking studies were .79 and .83 respectively. Sample items from the Hong scale include: "I consider advice from others to be an intrusion" and "I become frustrated when I am unable to make free and independent decisions," and "Advice and recommendations usually induce me to do just the opposite."

## Results

### *Effects of the Manipulations*

#### *Perceived threat to freedom*

In the flossing study, high threat participants perceived significantly stronger threat to freedom ( $M = 2.98$ ,  $SD = 1.07$ ) than did low threat participants ( $M = 2.31$ ,  $SD = 0.95$ ),  $F = 21.80$ ,  $df = 1, 200$ ,  $p < .001$ ,  $\eta = .32$ . Participants in the high threat condition of the drinking study perceived significantly stronger threat to freedom ( $M = 3.11$ ,  $SD = 1.16$ ) than participants in the low threat condition ( $M = 2.67$ ,  $SD = 0.98$ ),  $F = 8.67$ ,  $df = 1, 203$ ,  $p < .01$ ,  $\eta = .23$ . Both sets of results suggest that the threat to freedom manipulations were successful.

#### *Negative cognitive responses*

In the flossing study, high threat participants had significantly more negative thoughts ( $M = 1.76$ ,  $SD = 2.01$ ) than participants in the low threat condition ( $M = 1.19$ ,  $SD = 1.63$ ),  $F = 4.81$ ,  $df = 1, 200$ ,  $p < .05$ ,  $\eta = .20$ . Participants in the high threat condition of the drinking study had significantly more negative thoughts ( $M = 2.10$ ,  $SD = 2.02$ ) than participants in the low threat condition ( $M = 0.82$ ,  $SD = 1.18$ ),  $F = 29.58$ ,  $df = 1, 203$ ,  $p < .001$ ,  $\eta = .33$ .

#### *Anger*

In the flossing study, high threat participants experienced stronger anger ( $M = 0.90$ ,  $SD = 1.09$ ) than participants in the low threat condition ( $M = 0.45$ ,  $SD = 0.73$ ),  $F = 11.34$ ,  $df = 1, 200$ ,  $p < .01$ ,  $\eta = .21$ . Participants in the high threat condition of the drinking study reported significantly stronger anger ( $M = 1.44$ ,  $SD = 1.19$ ) than participants in the low threat condition ( $M = 0.70$ ,  $SD = 0.96$ ),  $F = 24.01$ ,  $df = 1, 203$ ,  $p < .001$ ,  $\eta = .34$ .

#### *Summary*

Results for the induction checks indicated that the threat manipulation successfully induced differences in perceived threat to freedom (as indexed by the four-item scale). Moreover, the threat manipulation produced differences in the number of negative cognitions and the intensity of anger in both data sets. These latter findings do not speak directly to questions regarding the nature of reactance. However, they do demonstrate that the data exhibit one of the necessary conditions for a meaningful test of mediation, that is, threat-induced variance in the proposed mediators.

### *Structural Equation Modeling*

#### *Input and model specification*

Tables 4 and 5 present the correlation matrix, means, and standard deviations of each variable from the drinking and flossing studies respectively. These summary data were used as input to LISREL 8.30 (Jöreskog & Sörbom, 1999). A covariance matrix was constructed, which was then used to estimate the parameters of the models using

**Table 4** Correlations, Means, and Standard Deviations for the Flossing Data ( $N = 196$ )

| Variable              | Mean               | SD                | 1     | 2     | 3     | 4     | 5     | 6    | 7    |
|-----------------------|--------------------|-------------------|-------|-------|-------|-------|-------|------|------|
| 1. Threat             | 0.00 <sup>a</sup>  | 1.06              | 1.00  |       |       |       |       |      |      |
| 2. Proneness          | 2.98 <sup>b</sup>  | 0.53 <sup>b</sup> | -0.03 | 1.00  |       |       |       |      |      |
| 3. Interaction        | -0.23 <sup>c</sup> | 7.35 <sup>c</sup> | 0.01  | -0.06 | 1.00  |       |       |      |      |
| 4. Anger              | 0.68               | 0.95              | 0.24  | 0.29  | 0.15  | 1.00  |       |      |      |
| 5. Negative cognition | 1.48               | 1.85              | 0.14  | 0.28  | 0.08  | 0.43  | 1.00  |      |      |
| 6. Attitude           | 4.66               | 0.49              | -0.11 | -0.15 | -0.11 | -0.22 | -0.27 | 1.00 |      |
| 7. Intention          | 7.21 <sup>b</sup>  | 3.00 <sup>b</sup> | 0.03  | -0.16 | -0.04 | -0.15 | -0.16 | 0.45 | 1.00 |

<sup>a</sup>High threat was coded as 1 and low threat as -1.

<sup>b</sup>Proneness was transformed by dividing HPRS score by 14 and intention was transformed by dividing response by 10 to minimize heteroscedasticity.

<sup>c</sup>The interaction term was mean centered.

maximum likelihood procedures. To reduce heteroscedasticity, the sum of the 14 Hong reactance proneness items was divided by 14 and the 100-point behavioral intention measure was divided by 10.

All of the variables in the models were treated as latent constructs<sup>2</sup>. Reactance proneness and attitude were corrected for measurement error by fixing the error term of the corresponding manifest variable at  $1-\alpha^2$  times its variance (Bollen, 1989). This procedure was followed for anger and negative cognitions in the Dual Process Model. However, these two variables were treated as manifest in the Intertwined Process Model and their measurement paths were estimated from the data. The measurement paths for threat and behavioral intention were set at 1.00.

#### *Criteria for evaluating the models*

Because the Dual Process Model contained the two Single Process alternatives, it was not necessary to run all four of the competing models. Rather, examination of the Dual

**Table 5** Correlations, Means, and Standard Deviations for the Drinking Data ( $N = 200$ )

| Variables             | Mean               | SD                | 1     | 2     | 3     | 4     | 5     | 6    | 7    |
|-----------------------|--------------------|-------------------|-------|-------|-------|-------|-------|------|------|
| 1. Threat             | 0.00 <sup>a</sup>  | 0.50              | 1.00  |       |       |       |       |      |      |
| 2. Proneness          | 2.97 <sup>b</sup>  | 0.58 <sup>a</sup> | -0.14 | 1.00  |       |       |       |      |      |
| 3. Interaction        | -1.13 <sup>c</sup> | 8.09 <sup>b</sup> | -0.01 | 0.05  | 1.00  |       |       |      |      |
| 4. Anger              | 1.06               | 1.14              | 0.32  | 0.15  | 0.01  | 1.00  |       |      |      |
| 5. Negative cognition | 1.47               | 1.76              | 0.37  | 0.16  | 0.09  | 0.45  | 1.00  |      |      |
| 6. Attitude           | 4.49               | 0.63              | 0.03  | -0.13 | 0.07  | -0.12 | -0.06 | 1.00 |      |
| 7. Intention          | 6.03 <sup>b</sup>  | 3.13 <sup>b</sup> | -0.04 | -0.14 | -0.01 | -0.27 | -0.24 | 0.06 | 1.00 |

<sup>a</sup>High threat was coded as 1 and low threat as -1.

<sup>b</sup>Proneness was transformed by dividing HPRS score by 14 and intention was transformed by dividing response by 10 to minimize heteroscedasticity.

<sup>c</sup>The interaction term was mean centered.

Process Model versus the Intertwined Process Model was sufficient to exhaust all logical possibilities. If the data supported the Single Process Cognitive Model, we should observe a significant path to and from negative cognition and non-significant paths to and from anger. The reverse pattern should obtain for the Single Process Affective Model.

For each model, strength of threat, reactance proneness, and a mean-deviated product term were specified as exogenous variables that influenced either affect and cognition (in the Dual Process Model) or a latent variable indexed by affect and cognition (in the Intertwined Process Model). These mediating variables were specified as the immediate causal antecedent of attitude, which in turn caused behavioral intention.

The models were evaluated on two criteria: significance of the path coefficients and overall fit. We planned to run each model, eliminate non-significant paths, and then repeat the analysis on the modified model with an eye toward changes in the goodness-of-fit indices. To evaluate the overall fit of the models to the data, we considered three fit indices. First, the Adjusted Goodness of Fit Index (AGFI) produces values ranging from 0 to 1 with values in excess of .90 indicating good fit. Second, Browne and Cudeck (1993) contend that values of the Root Mean Square Error of Approximation (RMSEA) of .08 or lower indicate reasonable fit, though values of .06 or below should be preferred. Third, the Bayesian Information Criterion (BIC; Raftery, 1995) is constructed such that negative values provide evidence of model fit, while positive BIC values suggest problematic model fit. One exceptional virtue of the BIC is its ability to compare the fit of non-nested models. Differences in BIC of 2 or more provide evidence favoring one model (with the smaller BIC value) over another; 6 or more provide strong evidence; and 10 is taken to be very strong evidence for model improvement (Raftery, 1995).

## Results

All of the predicted paths were significant in initial analyses of the flossing data. Consequently, we turned our attention to the fit indices (see Table 6), which uniformly preferred the Intertwined Process Model: AGFI (.94 vs. .87), RMSEA (.006 vs. .110), BIC (− 49.86 vs. − 16.49). The BIC difference of 33.37 was strong evidence in favor of the Intertwined Process Model. Figure 2 presents the path coefficients for the Intertwined Process Model.

Table 7 presents the fit indices for the drinking data, all of which favored the Intertwined Process Model over its competitors. That model had a larger AGFI (.93 vs. .76), a smaller RMSEA (0.077 vs. .180), and a smaller BIC (− 28.71 vs. 12.71). The BIC difference of 41.42

**Table 6** Fit Indices and Model comparison for the Flossing Data

| Models                    | <i>df</i> | AGFI <sup>a</sup> | RMSEA <sup>b</sup> | BIC <sup>c</sup> | BIC difference |
|---------------------------|-----------|-------------------|--------------------|------------------|----------------|
| Dual Process Model        | 9         | .87               | .110               | − 16.49          |                |
| Intertwined Process Model | 13        | .94               | .006               | − 49.86          | 33.37          |

<sup>a</sup>AGFI = Adjusted Goodness-of-fit Index.

<sup>b</sup>RMSEA = Root Mean Square Error Approximation.

<sup>c</sup>BIC = Bayesian Information Criterion.

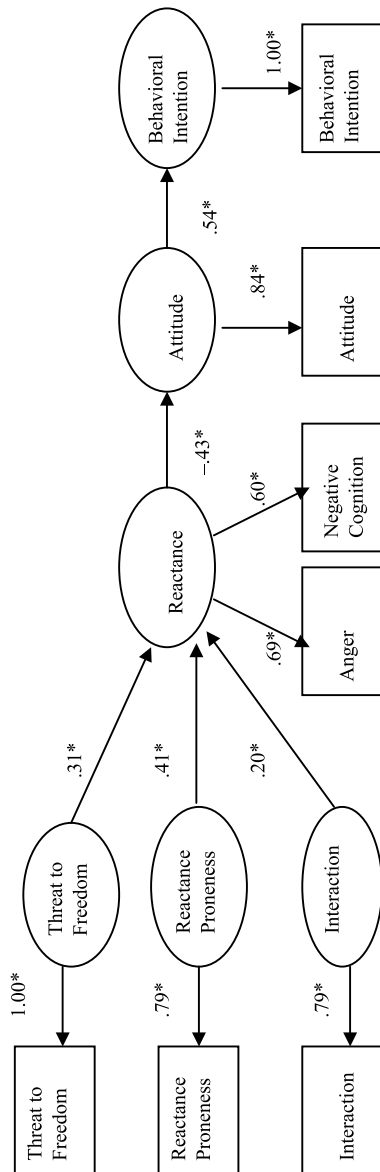


Figure 2 The obtained intertwined process model for the flossing data.

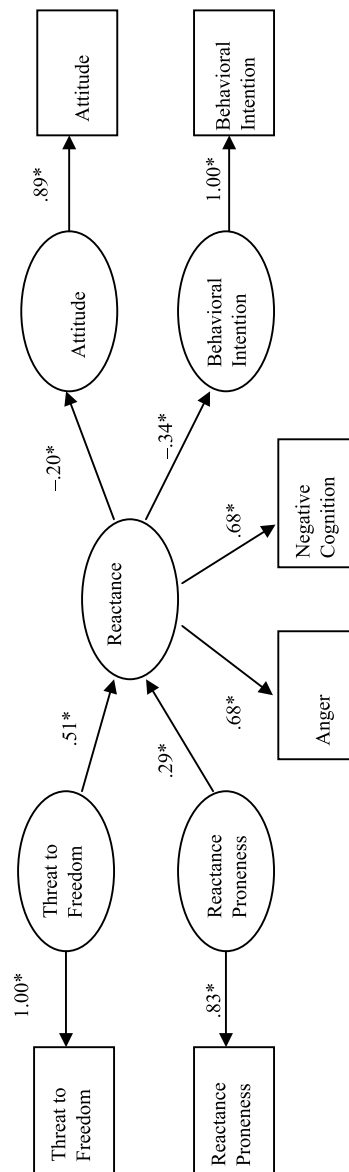


Figure 3 The obtained intertwined process model for the alcohol data.



also indicated the superiority of the model over the alternatives. However, analysis of the drinking data yielded non-significant paths for the interaction term onto reactance and for attitude onto intention. The interaction term was removed, first, and the remaining paths re-estimated. When the modification indices suggested the need for elimination of the attitude-intention link and the addition of a direct path from reactance to intention, we altered the model accordingly then re-estimated the parameters. Figure 3 presents the path coefficients for the final Intertwined Process Model.

#### *Research Questions 1 and 2: The Nature of Reactance*

The first two research questions inquired as to the nature of reactance. RQ1 asked: Is reactance best conceptualized as cognition, affect, or both? In fact, both data sets showed that cognition and affect mediated the effects of threat and proneness on attitude (see Tables 6 and 7). Thus, the two single process models were rejected.

RQ2 posed the question: If reactance has both cognitive and affective components, how are the two components combined? Here too, the results were consistent across the two data sets. The Intertwined Process Model was superior to the Dual Process Model, in terms of fit to the data, on both absolute (i.e., AGFI, RMSEA, BIC) and relative (BIC difference) indices (see Tables 6 and 7).

#### *Hypotheses 1 and 2: Antecedents of Reactance*

H1 predicted a positive association between threat to freedom and reactance. The path from threat to reactance was .31,  $p < .05$ , in the flossing data and .51,  $p < .05$ , in the drinking data. Therefore, Hypothesis 1 received consistent support across the two data sets.

H2 anticipated a positive correlation between reactance proneness and reactance. The path from proneness to reactance was .41,  $p < .05$ , in the flossing data and .29,  $p < .05$ , in the drinking data. Therefore, Hypothesis 2 was supported in both studies (see Figures 2 and 3).

#### *Research Question 3: Interaction*

RQ3 asked whether threat to freedom interacts with reactance proneness. The product term that represented this interaction was significant in the flossing data (.20,  $p < .05$ ), but not in the drinking data (.05, *ns*). Thus, the two studies did not provide a consistent answer to RQ3 (see Figures 2 and 3). Regression analysis was used to explore the functional form of the interaction in the flossing data. The data were broken on threat, and the composite measure of reactance, using standardized measures of anger and negative cognition, was regressed on reactance proneness. In the high threat condition, the analysis yielded  $r = .40$ ,  $p < .001$ , whereas in the low threat condition, the coefficient was  $r = .27$ ,  $p < .001$ . That the interaction term was significant indicated that the two coefficients were themselves significantly different. These results showed that reactance was strongest when both threat and reactance proneness were high, and weakest when both were low. Thus, the interaction was of a form that was compatible with the theory.

**Table 7** Fit Indices and Model Comparison for the Drinking Data

| Models                          | <i>df</i> | AGFI <sup>a</sup> | RMSEA <sup>b</sup> | BIC <sup>c</sup> | BIC difference |
|---------------------------------|-----------|-------------------|--------------------|------------------|----------------|
| Dual Process Model              | 9         | .77               | .161               | 5.65             |                |
| Intertwined Model               | 11        | .90               | .089               | − 28.14          | 33.79          |
| Final Intertwined Process Model | 9         | .93               | .077               | − 28.71          |                |

<sup>a</sup>AGFI = Adjusted Goodness-of-fit Index.<sup>b</sup>RMSEA = Root Mean Square Error Approximation.<sup>c</sup>BIC = Bayesian Information Criterion.

## Discussion

Our primary concern in this research was to illuminate the nature of reactance and the operation in the persuasion process. However, because of the important ramifications of those theoretical questions for applied undertakings, we chose to test our thinking in the context provided by two health issues: flossing and drinking. In the remainder of this paper, we first examine the conceptual issues, then turn our attention to questions of application.

### *The Nature of Reactance*

Brehm and Brehm (1981) argued that reactance could not be measured: The only means available for assessing the presence of reactance was to infer it from its effects. Our results, instead, suggest that it is possible to use a combination of self-report cognitive and emotional measures to create a more or less direct index of reactance. This conclusion rests on the ability of the combined variables to mediate the effects of threat and reactance proneness on attitude and behavioral intention. However, there are limitations to these conclusions that arise from the design of our experiment. First, among them is the fact that our project demonstrated mediation for only two independent variables and for only two message topics. Additional tests would strengthen our certainty about the generalizability of the results.

The data clearly favored the Intertwined Process Model, which claimed that reactance is best understood as an intermingling of negative cognition and anger. Moreover, because the factor loadings for anger and cognition on reactance are similar in magnitude in both data sets, it appears that each contributes about equally to the motivation to restore freedom. But our claim that affect and cognition constitute a psychological alloy is also limited by aspects of our research design. Consider that cognition and affect are phenomena capable of rapid change. For example, an appropriate time frame for the retrieval and integration of some elements of thought may be on the order of 100 ms or less (van Turennout, Hagoort, & Brown, 1998). In contrast, our data aggregated thoughts and feelings over the time that respondents took to read the recommendation components of the two messages (i.e., the last page of six-page messages). Informal observation indicated that this was a period of 10–15 s. Thus, the data do support the idea that cognitive and affective responses that characterize

reactance are empirically inseparable, but that claim is limited to the 10–15 s period assessed in this study. Although this in no way compromises the conclusion, it does suggest that researchers attempting to model cognition and affect as indicators of reactance over an entire message may not be able to replicate our results.

### *Antecedents of Reactance*

A great deal of research has been conducted on reactance theory since the appearance of Brehm's (1966) treatise. Several studies have claimed to examine the effects of threat and reactance proneness on reactance (e.g., Doob & Zabrack, 1971; Heller et al., 1973; Kohn & Barnes, 1977; Worchel & Brehm, 1970). In reality, however, those projects have tested the impact of threat and proneness on some outcome variable such as attitude and behavior. Though cast as a mediator, reactance went unmeasured. We believe that our data are the first to demonstrate an effect for these two variables on reactance itself, that is, the combination of negative cognition and anger. In that respect, the results make a unique contribution to literature on reactance theory.

We anticipated that message variables that enhanced the strength of the threat to freedom would show a positive association and this proved to be the case. Although our findings are consistent with previous findings (e.g., Benoit, 1998; Bensley & Wu, 1991; Doob & Zabrack, 1971; Heller et al., 1973; Kohn & Barnes, 1977; Worchel & Brehm, 1970), one might critique our messages on two grounds. One is that we knowingly confounded intent to persuade with language intensity. Another is that we may have unwittingly manipulated other message variables. For example, one reviewer noted that both low-threat messages mention specific groups (i.e., "The majority of dentists view that evidence as strong and clear" and "Most people agree that over-consumption of alcohol is a serious campus problem"). From our perspective, neither of these issues poses a concern because the aim of the study was not to generate knowledge about the effects of specific message features on reactance. Instead, we sought develop strong inductions and, importantly, to create messages that resembled those used in previous research. This latter point was an over-riding concern. In fact, the credibility of the tests of mediation hinged on independent and dependent variables that were derived from the reactance literature in obvious ways. This was true for the message variables as well as the measure of trait reactance. The second hypothesis suggested that individuals who are prone to reactance would experience a higher level of reactance as a function of the messages than persons who were not so inclined. On this point too, the data supported our expectations, and in so doing, lend further credence to claims of mediation.

RQ3 asked about the existence and the form of the interaction between threat to freedom and trait reactance proneness. The data showed a significant interaction in the flossing data that was of a form compatible with the theory. However, the interaction effect was not replicated in the alcohol data. Taking these two findings together, we can only conclude that reactance proneness may interact with threat for some topics and not for others. To appreciate the nature of this conclusion, it is important to bear in mind that trait reactance proneness represents a propensity to

experience a state, not the state itself. Even individuals who are high in trait proneness are not constantly in state of reactance. Rather, they respond more strongly to the same stimuli than do persons low in trait proneness. Our data show that this tendency is additive for one topic and multiplicative for another, thereby suggesting that some topics are especially potent with regard to their ability to elicit state reactance from individuals high in the propensity to experience reactance.

One implication of these findings is that message designers should avoid high threat messages altogether. While we would endorse that reasoning, it is important to remember that threat is not a property of messages, but rather a judgment made by members of the target audience. Thus, advice to message designers to cast messages as informative, rather than persuasive, and to steer clear of dogmatic language is sound, but limited. A course of action that might produce more valuable knowledge would be to work toward a theoretically grounded means of classifying health topics that are likely to interact with strength of threat versus those that are not. Reactance theory and research directs our attention to variables such as importance of the behavior (Bensley & Wu, 1991) or prior intention (Buller et al., 2000). But other concepts may need to be considered too. The two topics examined in this research differed on the extent to which the behavior is public versus private as well as the direction of the behavior change that was called for (i.e., *more flossing* vs. *less drinking*).

#### *Implications for Assessing Reactance Induced by Persuasive Messages*

Although we stand by the conclusion that reactance should be viewed as a cognitive and affective amalgam, it does not follow that all anger or all negative cognitions result from perceived threats to freedom. In fact, from a theoretical standpoint, we prefer to view reactance simply as the output of more general and frequently interdependent psychological systems (i.e., cognition and emotion). In practice, the tight interweaving of cognition and affect assumes less significance. As the results relate to the evaluation of persuasive health messages, we see two important conclusions: (a) Reactance can be measured using well-known and commonly applied self-report techniques such as those used in this study, and (b) both anger and negative cognitions should be assessed. For purposes of formative or summative message evaluation, it matters little whether or not a composite is created from them. As long as both are assessed the evaluation will be comprehensive. In fact, as we noted earlier, efforts to treat affect and cognition as indicators of a latent construct may fail in research designs that focus on whole messages rather than message components. Evaluation researchers should take heart from the fact that they already have in hand the tools to assess reactance as well as other affects and cognitions.

#### *Reactance Theory and Message Design*

Further inquiry is warranted into the question of how messages can be designed that will defuse, or at least not exacerbate, innate tendencies toward reactance.

As this research proceeds, it will be important to acknowledge the limits of the relatively narrow view of communication that is characteristic of inquiry in the reactance tradition. As was done in the current investigation, studies of reactance typically characterize persuasive messages uni-dimensionally in terms of degree of threat. However, empirical research on the perception of influence messages clearly reveals that individuals utilize at least three distinct dimensions (see Dillard, Wilson, Tusing, and Kinney, 1997, for a summary). *Explicitness* is the degree to which the language of the message makes plain the source's intent. Roughly parallel to forcefulness or authoritarianism, *dominance* captures the extent to which a message reveals that the source believes he or she can control the message recipient. Finally, *reason* is present in any given message when justifications are offered in support of the claim that audience members should adopt the position advocated by the source. Although our knowledge base concerning the effects of these three dimensions is not large, such data as do exist suggest that dominance causes anger (Dillard, Kinney, & Cruz, 1996), whereas reason giving softens perceptions of intrusiveness (Dillard et al., 1997) and is, therefore, less likely to provoke anger/reactance. The effect of explicitness is likely to be highly context dependent, but the available research shows that it generally enhances persuasion (O'Keefe, 1997) and may produce positive emotional responses (Dillard et al., 1996). In light of the distinctiveness and variability of the impact of these aspects of message perception, it is essential that message design research move beyond a unidimensional treatment of threat.

#### *Attitude-Behavior Correspondence: Flossing, Binge Drinking, and Beyond*

Some very well-established theories cast attitude as the immediate causal precursor of behavioral intention (Ajzen, 1991; Fishbein & Ajzen, 1975). In this project, that theoretical sequence was supported in the flossing data, but not in the drinking study. In fact, although both attitude toward reducing alcohol consumption and intention to reduce consumption were negatively influenced by reactance, the two outcome variables were empirically independent of one another in the alcohol study. We believe that the findings for the two specific topics may be illustrative of two more general cases.

For flossing, attitude and intention correspond well with one another because individuals approach the topic with only one aim in mind, that is, health maintenance. They consider the relevant facts, integrate that information to form an attitude, and then generate a corresponding intention. In other words, they behave in a reasoned, planful manner. Accordingly, the prospects are bright for communication campaigns that adopt a rational approach to increasing the frequency of dental flossing. For those topics about which individuals are concerned *only* with self-preservation, the likelihood of success seems high for appeals that detail the costs and benefits of behavior change.

Persuasive messages concerning alcohol are likely to be processed quite differently from appeals to increase dental flossing. Whereas message recipients still may be

responsive to arguments to protect their health, they are likely to be simultaneously influenced by an awareness of the many social and commercial forces that promote binge drinking on college campuses. When these forces are strong in the aggregate, message recipients may believe that actual behavior is beyond their control. In fact, Wall, Hinson, and McKee (1998) report an inverse relationship between perceived control over excessive drinking and intention to drink to excess. Under such circumstances, attitude toward the behavior is functionally irrelevant to the intention itself (Greenwald, 1989). The result is an empirical disconnect between the two constructs: The attitude becomes unrelated to action because behavior is the result of other forces such as perceived norms (Fishbein & Yzer, 2003). In these instances, a “successful” persuasive message might change attitude without any corresponding impact on intention. However, as the alcohol data show (see Figure 3), the aftermath of reactance arousal is less benign. It can exert a direct effect on behavioral intention. This is a particularly worrisome problem for health campaigns in light of the fact that those persons at greatest risk are often the same persons for whom the behavior is most important. An illustration of this point can be found in Bensley and Wu (1991), a study in which the importance of drinking (indexed by prior behavior) interacted with strength of threat to produce substantially heightened alcohol consumption among male, heavy drinkers. In combination with the current project, such findings underscore the difficulty of conducting any campaign on a general audience without arousing reactance among members of the target group. Not to be overlooked is the possibility of social and cultural iatrogenic effects that may have their roots in reactance (Guttman, 2000). Of course, such pessimistic possibilities do not mean that every effort at enhancing public health will do harm. Well-constructed messages that are subjectively evaluated as compelling have the potential to persuade even in the presence of reactance (Brehm, 1966; Brehm & Brehm, 1981). Surely some consideration of the balance between good and ill effects is needed at the planning and evaluation stages of every persuasive campaign.

## Summary

Two studies were carried out to explore the nature of reactance and its role in the impact of persuasive health messages. Whereas the theory itself contends that reactance cannot be measured, this project supported a different conclusion. The data indicated that reactance might reasonably be conceived of as an amalgam of anger and negative cognitions. This result is useful to efforts to improve public health in that reactance can be assessed in formative and evaluative research using well-known and widely understood self-report methods. The study also suggested that strength of threat may interact with trait reactance proneness, but that it is unlikely to do so for all health topics. Finally, the data implied that attitude-behavior correspondence may be strong for certain types of health behaviors and weak for others. We speculated that conditions that place the goal of self-preservation in conflict with other motives might help to explain poor correspondence. In all, these

results extend our understanding of reactance theory and its application to persuasive health communication.

## Notes

- [1] One alternative to this position is to conceptualize reactance in other terms. For example, as one anonymous reviewer pointed out, one might think of reactance as a cognitive appraisal of restriction or illegitimacy. If so, self-report would presumably provide a valid means of measuring reactance. We concur with this reasoning, but note (as the reviewer did as well) that it would require one to explicate how the appraisal differs from counter-arguing. More to the point, we think that to treat reactance as a purely cognitive judgment is to rob it of the motivational properties that Brehm so clearly believed it possesses. Hence, while we admit to the logical possibility of conceptualizing reactance in different terms, we see the solution as one that creates substantial distance from the original theory and judge it undesirable for that reason.
- [2] Some readers might reasonably object to our models on the grounds that we did not treat the perceived threat induction check as a mediator between the induction and mediators of interest (i.e., affect and cognition). Although we believe that such an analysis is desirable in principle, it would have precluded tests of the interaction between strength of threat and reactance proneness (because one cannot form an interaction term between an endogeneous and an exogeneous variable). Because the interaction was something that we wished to test, we opted for models that allowed us to do that.

## References

- Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50, 179–211.
- Benoit, W. L. (1998). Forewarning and persuasion. In M. Allen, & R. W. Preiss (Eds.), *Persuasion: Advances through meta-analysis* (pp. 139–154). Cresskill, NJ: Hampton Press Inc.
- Bensley, L. S., & Wu, R. (1991). The role of psychological reactance in drinking following alcohol prevention messages. *Journal of Applied Social Psychology*, 21, 1111–1124.
- Bollen, K. (1989). *Structural equations with latent variables*. New York: Wiley.
- Brehm, J. W. (1966). *A theory of psychological reactance*. New York: Academic Press.
- Brehm, J. W., Stires, L. K., Sensenig, J., & Shaban, J. (1966). The attractiveness of an eliminated choice alternative. *Journal of Experimental Social Psychology*, 2, 301–313.
- Brehm, S. S., & Brehm, J. W. (1981). *Psychological reactance: A theory of freedom and control*. New York: Academic Press.
- Brown, M. W., & Cudeck, R. (1993). Alternative ways of assessing model fit. In K. A. Bollen, & J. S. Long (Eds.), *Testing structural equation models* (pp. 136–162). Newbury Park, CA: Sage.
- Buller, D. B., Borland, R., & Burgoon, M. (1998). Impact of behavioral intention on effectiveness of message features: Evidence from the family sun safety project. *Human Communication Research*, 24, 422–453.
- Buller, D. B., Burgoon, M., Hall, J. R., Levine, N., Taylor, A. N., Beach, B., Buller, M. K., & Melcher, C. (2000). Long-term effects of language intensity in preventive messages on planned family solar protection. *Health Communication*, 12, 262–275.
- Burgoon, M., Alvaro, E., Grandpre, J., & Voulodakis, M. (2002). Revisiting the theory of psychological reactance: Communicating threats to attitudinal freedom. In J. P. Dillard, & M. W. Pfau (Eds.), *The persuasion handbook: Developments in theory and practice* (pp. 213–232). Thousand Oaks, CA: Sage.



- Cacioppo, J. T., & Petty, R. E. (1981). Social psychological procedures for cognitive response assessment: The thought-listing technique. In T. V. Merluzzi, C. R. Glass, & M. Genest (Eds.), *Cognitive assessment* (pp. 309–342). New York: Guilford Press.
- Chaffee, S. H. (1991). *Explication*. Newbury Park, CA: Sage.
- Derryberry, D., & Tucker, D. M. (1994). Motivating the focus of attention. In P. N. Niedenthal, & S. Kitayama (Eds.), *The heart's eye: Emotional influences in perception and attention* (pp. 167–196). San Diego, CA: Academic Press.
- Dillard, J. P., Kinney, T. A., & Cruz, M. G. (1996). Influence, appraisals, and emotions in close relationships. *Communication Monographs*, 63, 105–130.
- Dillard, J. P., & Meijnders, A. (2002). Persuasion and the structure of affect. In J. P. Dillard, & M. Pfau (Eds.), *The persuasion handbook: Developments in theory and practice* (pp. 309–327). Newbury Park, CA: Sage.
- Dillard, J. P., & Peck, E. (2000). Affect and persuasion: Emotional responses to public service announcements. *Communication Research*, 27, 461–495.
- Dillard, J. P., & Peck, E. (2001). Persuasion and the structure of affect: Dual systems and discrete emotions as complementary models. *Human Communication Research*, 27, 38–68.
- Dillard, J. P., Plotnick, C. A., Godbold, L. C., Freimuth, V. S., & Edgar, T. (1996). The multiple affective consequences of AIDS PSAs: Fear appeals do more than scare people. *Communication Research*, 23, 44–72.
- Dillard, J. P., Wilson, S. R., Tusing, K. J., & Kinney, T. A. (1997). Politeness judgments in personal relationships. *Journal of Language and Social Psychology*, 16, 297–325.
- Doob, A. N., & Zabrack, M. (1971). The effect of freedom-threatening instructions and monetary inducement on compliance. *Canadian Journal of Behavioral Science*, 3, 408–412.
- Dowd, E. T., Hughs, S. L., Brockbank, L., & Halpain, D. (1988). Compliance-based and defiance-based intervention strategies and psychological reactance in the treatment of free and unfree behavior. *Journal of Counseling Psychology*, 35, 370–376.
- Fishbein, M., & Ajzen, I. (1975). *Belief, attitude, intention, and behavior: An introduction to theory and research*. Reading, MA: Addison-Wesley.
- Fishbein, M., & Yzer, M. C. (2003). Using theory to design effective health behavior interventions. *Communication Theory*, 13, 164–183.
- Foxcraft, D. R., Lister-Sharp, D., & Lowe, G. (1997). Alcohol misuse prevention for young people: A systematic review reveals methodological concerns and lack of reliable evidence of effectiveness. *Addiction*, 92, 531–537.
- Greenwald, A. G. (1989). Why are attitudes important? In A. R. Pratkanis, S. J. Breckler, & A. G. Greenwald (Eds.), *Attitude structure and function* (pp. 1–11). Hillsdale, NJ: Lawrence Erlbaum Associates, Inc.
- Guttman, N. (2000). *Public health communication interventions: Values and ethical dilemmas*. Thousand Oaks, CA: Sage.
- Guttman, N., Kegler, M., & McLeroy, K. R. (1996). Health promotion paradoxes, antinomies, and conundrums. *Health Education Research, Theory, and Practice*, 11, i–xiii.
- Hammock, T., & Brehm, J. W. (1966). The attractiveness of choice alternatives when freedom to choose is eliminated by a social agent. *Journal of Personality*, 34, 546–554.
- Heller, J., Pallak, M., & Picek, J. (1973). The interactive effects of intent and threat on boomerang attitude change. *Journal of Personality and Social Psychology*, 26, 273–279.
- Hong, S. M. (1992). Hong's Psychological Reactance Scale: A further factor analytic validation. *Psychological Reports*, 70, 512–514.
- Hong, S. M., & Faedda, S. (1996). Refinement of the Hong Psychological Reactance Scale. *Educational and Psychological Measurement*, 56, 173–182.
- Hong, S. M., & Page, S. (1989). A psychological reactance scale: Development, factor structure and reliability. *Psychological Reports*, 64, 1323–1326.



- Hornik, R. (2002). Public health communication: Making sense of the evidence. In R. Hornik (Ed.), *Public health communication* (pp. 1–19). Mahwah, NJ: Lawrence Erlbaum Associates, Inc.
- Jöreskog, K., & Sörbom, D. (1996). *Lisrel 8: User's reference guide*. Chicago: Scientific Software International.
- Kelly, A. E., & Nauta, M. M. (1997). Reactance and thought suppression. *Personality and Social Psychology Bulletin*, 23, 1123–1132.
- Kim, K. S., & Hunter, J. E. (1993a). Attitude–behavior relations: A meta-analysis of attitudinal relevance and topic. *Journal of Communication*, 43, 101–142.
- Kim, K. S., & Hunter, J. E. (1993b). Relationships among attitudes, behavioral intentions, and behavior. *Communication Research*, 20, 331–364.
- Kohn, P. M., & Barnes, G. E. (1977). Subject variables and reactance to persuasive communications about drugs. *European Journal of Social Psychology*, 7, 97–109.
- Kuhl, J. (1986). Motivation and information processing: A new look at decision making, dynamic change, and action control. In R. M. Sorrentino, & E. T. Higgins (Eds.), *Handbook of motivation and cognition: Foundations of social behavior* (pp. 404–434). New York: John Wiley.
- Lang, P. J. (2000). Emotion and motivation: Attention, perception, and action. *Journal of Sports and Exercise Psychology*, 22, S122–S140.
- Leventhal, H. (1970). Findings and theory in the study of fear communications. In L. Berkowitz (Ed.), *Advances in experimental social psychology*, (Vol. 5, pp. 119–186). New York: Academic Press.
- Merz, J. (1983). A questionnaire for the measurement of psychological reactance [in German]. *Diagnostica*, 29, 75–82.
- Nabi, R. L. (2002). Discrete emotions and persuasion. In J. P. Dillard, & M. Pfau (Eds.), *The persuasion handbook: Developments in theory and practice* (pp. 289–308). Newbury Park, CA: Sage.
- O'Keefe, D. J. (1997). Argumentative candor and persuasive success: A meta-analysis of the persuasive effects of implicit and explicit message conclusions. In J. F. Klumpp (Ed.), *Argument in a time of change: Definitions, frameworks, and critiques* (pp. 63–69). Annandale, VA: National Communication Association.
- Petty, R. E., & Cacioppo, J. T. (1986). *Communication and persuasion: Central and peripheral routes to attitude change*. New York: Springer-Verlag.
- Raftery, A. E. (1995). Bayesian model selection in social research. In P. V. Marsden (Ed.), *Sociological methodology* (pp. 111–163). Cambridge, UK: Basil Blackwell.
- Ringold, D. J. (2002). Boomerang effect in response to public health interventions: Some unintended consequences in the alcoholic beverage market. *Journal of Consumer Policy*, 25, 27–63.
- Rogers, R. W. (1983). Cognitive and physiological processes in fear appeals and attitude change: A revised theory of protection motivation. In J. Cacioppo, & R. Petty (Eds.), *Social psychophysiology* (pp. 153–176). New York: Guilford Press.
- Schwarz, N., Frey, D., & Kumpf, M. (1980). Interactive effects of writing and reading a persuasive essay on attitude change and selective exposure. *Journal of Experimental Social Psychology*, 16, 1–17.
- Seltzer, L. F. (1983). Influencing the “shape” of resistance: An experimental exploration of paradoxical directives and psychological reactance. *Basic and Applied Social Psychology*, 4, 47–71.
- Shaver, P., Schwartz, J., Kirson, D., & O'Connor, C. (1987). Emotion knowledge: Further explorations of a prototype approach. *Journal of Personality and Social Psychology*, 52, 1061–1086.
- Shen, L., & Dillard, J. P. (in press). Psychometric properties of the Hong Psychological Reactance Scale. *Journal of Personality Assessment*.
- Smith, M. J. (1977). The effects of threats to attitudinal freedom as a function of message quality and initial receiver attitude. *Communication Monographs*, 44, 196–206.

- Stephenson, M. T. (2003). Examining adolescents' responses to antimarijuana PSAs. *Human Communication Research*, 29, 343–369.
- Stewart, D. W., & Martin, I. M. (1994). Intended and unintended consequences of warning messages: A review and synthesis of empirical research. *Journal of Public Policy and Marketing*, 13, 1–19.
- Van Turenout, M., Hagoort, P., & Brown, C. M. (1998). Brain activity during speaking: From syntax to phonology in 40 milliseconds. *Science*, 280, 572–574.
- Wall, A. -M., Hinson, R. E., & McKee, S. A. (1998). Alcohol outcome expectancies, attitudes toward drinking and the theory of planned behavior. *Journal of Studies on Alcohol*, 59, 409–419.
- Wallack, L. M. (1981). Mass media campaigns: The odds against finding behavior change. *Health Education Quarterly*, 8, 209–260.
- White, G. L., & Zimbardo, P. G. (1980). The effects of threat of surveillance and actual surveillance on expressed opinion toward marijuana. *Journal of Social Psychology*, 111, 49–61.
- Wicklund, R. A. (1974). *Freedom and reactance*. Potomac, MD: Lawrence Erlbaum Associates, Inc.
- Wilde, G. J. S. (1993). Effects of mass media communications on health and safety habits: An overview of issues and evidence. *Addiction*, 88, 983–996.
- Witte, K. (1994). Fear control and danger control: A test of the extended parallel process model (EPPM). *Communication Monographs*, 61, 113–134.
- Worchel, S. (1974). The effect of three types of arbitrary thwarting on the instigation to aggression. *Journal of Personality*, 42, 300–318.
- Worchel, S., & Andreoli, V. (1974). Attribution of causality as a means of restoring behavioral freedom. *Journal of Personality and Social Psychology*, 29, 237–245.
- Worchel, S., Andreoli, V. A., & Archer, R. (1976). When is a favor a threat to freedom: The effects of attribution and importance of freedom on reciprocity. *Journal of Personality*, 44, 294–310.
- Worchel, S., & Brehm, J. (1970). Effect of threats to attitudinal freedom as a function of agreement with the communicator. *Journal of Personality and Social Psychology*, 14, 18–22.